UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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| COURSE | INTRODUCTION TO INTERACTIVE MULTIMEDIA |
| COURSE CODE | CSC413 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 5 printed pages.*

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## PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

**1.** Authoring tools help developers organise and edit multimedia elements.
**2.** A URL identifies the address of a web resource.
**3.** HTML is used to structure content on web pages.
**4.** RGB is an additive colour model.
**5.** Sampling converts a continuous sound wave into digital values.
**6.** Interlacing displays every line of a video frame at the same time.
**7.** A scanner is an input device.
**8.** Streaming allows media to begin playing before the whole file is downloaded.
**9.** A prototype is an early model of a product.
**10.** Tweening creates intermediate frames between keyframes.
**11.** Interactivity gives users control over content and flow.
**12.** Higher resolution always means a smaller file.
**13.** A plotter is an input device.
**14.** The amplitude of a sound wave determines its loudness.
**15.** A higher sampling rate generally produces a more accurate recording.

## PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

**Table A**

| | | | | |
|---|---|---|---|---|
| A. Keyframe | B. MIDI | C. Bitmap | D. Target audience | E. Multimedia |
| F. Hypermedia | G. Virtual reality | H. Storyboard | I. Linear multimedia | J. Interactivity |
| K. Serif | L. Convergence | M. Card-based tool | N. Scanner | O. Sampling |

**1.** image made of pixels
**2.** simulated environment
**3.** visual sequence plan
**4.** converting sound into digital values
**5.** font with small strokes at character ends
**6.** authoring tool organised like pages or cards
**7.** intended users of a product
**8.** integration of text, graphics, audio, video and animation
**9.** presentation without user control of sequence
**10.** standard for musical notes and instructions
**11.** linked multimedia elements
**12.** user control over content and flow
**13.** merging of computing, media and communications
**14.** input device that digitises documents
**15.** important frame in an animation

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## PART C (30 MARKS)

Choose the correct answer for each question.

**1.** A browser is used to:
A. Print documents only
B. Scan images
C. Display web pages
D. Edit audio

**2.** Resolution refers to:
A. Number of pixels in an image
B. File type
C. Font size
D. Colour name

**3.** Which is a benefit of multimedia in education?
A. It removes all interaction
B. It uses only text
C. It replaces all teachers
D. It combines visuals and sound

**4.** Hypertext contains:
A. Only images
B. No text
C. Links to other content
D. Only audio

**5.** Which is a 3D animation task?
A. Kerning
B. Modelling and rendering
C. Sampling
D. Bookmarking

**6.** A licence gives:
A. Ownership of all works
B. Permission to use a work
C. Free hardware
D. Unlimited bandwidth

**7.** Path animation moves an object:
A. Along a predefined route
B. Randomly
C. Without motion
D. Only vertically

**8.** Cel animation is created by:
A. Scanning text
B. Recording sound
C. Printing images
D. Drawing each frame by hand

**9.** RGB is used mainly for:
A. Printed output
B. Audio recording
C. Video editing only
D. Screen display

**10.** Streaming allows media to:
A. Require full download first
B. Only play offline
C. Play while downloading
D. Remove compression

**11.** Which is NOT a multimedia element?
A. Text
B. Compiler
C. Audio
D. Animation

**12.** Kerning adjusts:
A. Space between individual character pairs
B. Font colour
C. Line length
D. Page size

**13.** A prototype is:
A. A font
B. An early model of a product
C. A final release
D. A codec

**14.** Card-based authoring tools are suited to:
A. Audio editing only
B. Messages with a beginning and end
C. Live video only
D. 3D modelling only

**15.** Tweening creates:
A. Sound effects
B. Text fonts
C. In-between frames
D. Video codecs

**16.** A storyboard is:
A. A visual sequence plan
B. A sound file
C. A video codec
D. A colour model

**17.** Which is an input device?

A. Monitor
B. Scanner
C. Printer
D. Plotter

**18.** Hypermedia is best described as:

A. Multimedia with hyperlinks
B. A single video file
C. Text-only media
D. A printed manual

**19.** Interlacing:

A. Draws every line at once
B. Draws alternate lines of a frame
C. Adds sound
D. Removes colour

**20.** Keyframes are:

A. Colour profiles
B. Important frames in an animation
C. Audio samples
D. Web links

**21.** Which colour model is subtractive?

A. RGB
B. HSL
C. HSV
D. CMYK

**22.** Chroma key is used to:

A. Compress audio
B. Add text
C. Scan images
D. Replace a colour background

**23.** Which format is commonly used for bitmap images?

A. MPEG
B. PNG
C. HTML
D. MIDI

**24.** A bitmap image is made of:

A. Sound samples
B. Text characters
C. Pixels
D. Mathematical objects

**25.** MIDI stores:

A. Musical notes and instructions
B. Digitized audio samples
C. Images
D. Video frames

**26.** RAM is:

A. An input device
B. A web browser
C. Temporary working memory
D. Permanent storage

**27.** MPEG is:

A. An image editor
B. A storage device
C. A video compression family
D. A font

**28.** Frame rate is:

A. Frames per second
B. Bits per second
C. Pixels per inch
D. Samples per second

**29.** Which is an output device?

A. Plotter
B. Scanner
C. Keyboard
D. Mouse

**30.** Which is an example of linear multimedia?

A. An interactive kiosk
B. A website with menus
C. A game with choices
D. A cinema presentation

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## PART D (40 MARKS)

### QUESTION 1

A company wants an interactive product demonstration that runs on a kiosk and on the web.

**a)** State TWO features to consider when choosing authoring software.

*(2 marks)*

**b)** Explain the interactivity level suitable for the demonstration.

*(4 marks)*

**c)** Explain TWO authoring tool categories appropriate for this project.

*(4 marks)*

### QUESTION 2

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

**a)** State TWO animation file formats suitable for this project.

*(3 marks)*

**b)** Describe the steps of cel animation and ONE advantage of using it.

*(3 marks)*

**c)** Differentiate 2-D and 2.5-D animation with ONE example for each.

*(4 marks)*

### QUESTION 3

A museum wants a website that presents its collection and lets visitors explore related exhibits.

**a)** State TWO differences between linear and non-linear navigation.

*(2 marks)*

**b)** Describe TWO roles of hyperlinks in the website.

*(3 marks)*

**c)** Explain how streaming and compression affect multimedia delivery on the web.

*(5 marks)*

### QUESTION 4

A shopping complex requires a kiosk that gives customers location and retailer information.

**a)** State TWO important project team roles for completing the kiosk.

*(3 marks)*

**b)** Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

*(3 marks)*

**c)** Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

*(4 marks)*

*END OF QUESTION PAPER*